time up so strongly, as seen in all reference frames, that it produces, at least briefly, a time machine.

We also know that, if an infinitely advanced civilization somehow acquires a wormhole, then the only way to hold the wormhole open (so it can be used for interstellar travel) is by threading it with exotic material. We know that vacuum fluctuations of the electromagnetic field are a promising form of exotic material: They can be exotic (have a negative average energy density as measured by a light beam) in curved spacetime under a wide variety of circumstances. However, we do *not* yet know whether they can be exotic inside a wormhole and thereby hold the wormhole open.

In the pages to come, I shall assume that an infinitely advanced civilization has somehow acquired a wormhole and is holding it open by means of some sort of exotic material; and I shall ask what other uses, besides interstellar travel, the civilization might find for its wormhole.

Time Machines

In December 1986, the fourteenth semi-annual Texas Symposium on Relativistic Astrophysics was held in Chicago, Illinois. These "Texas" symposia, patterned after the 1963 one in Dallas, Texas, where the mystery of quasars was first discussed (Chapters 7 and 9), had by now become a firmly established institution. I went to the symposium and lectured on dreams and plans for LIGO (Chapter 10). Mike Morris (my "wormhole" student) also went, to get his first full-blown exposure to the international community of relativity physicists and astrophysicists.

In the corridors between lectures, Morris became acquainted with Tom Roman, a young assistant professor from Central Connecticut State University who, several years earlier, had produced deep insights about exotic matter. Their conversation quickly turned to wormholes. "If a wormhole can really be held open, then it will permit one to travel over interstellar distances far faster than light," Roman noted.

"Doesn't this mean that one can also use a wormhole to travel backward in time?"

How stupid Mike and 1 felt! Of course; Roman was right. We, in fact, had learned about such time travel in our childhoods from a famous limerick: